AQ9.6: Activity Questions 6 - Not Graded

This assignment will not be graded and is only for practice.

Level 1

1 point

Consider a function $f: R^3 \rightarrow R$f : R3 → R.

Choose the correct statement with respect to ∇f∇f.

- [ ] ∇f∇f has co-domain R^2 R2
- [ ] ∇f∇f has co-domain RR
- [ ] ∇f∇f has co-domain R^3 R3
- [ ] ∇f∇f has domain R^3 R3

No, the answer is incorrect.

Score: 0

Accepted Answers:

∇f∇f has co-domain R^3 R3

∇f∇f has domain R^3 R3

1 point

Let $f: R^3 \rightarrow R$f : R3 → R and suppose its gradient ∇f∇f is continuous. Which of the following statements is true for the directional derivative of ff in the direction of $a \in R^3$ a ∈ R3 ? Consider dot product to be the inner product $< , >$ <, >.

- ○ $D_a f = \frac{1}{\| a \|} < a, \nabla f >$ Daf = ‖a‖1 < a, ∇f >
- ○ $D_a f = < a, \nabla f >$ Daf =< a, ∇f >
- ○ $D_a f = \frac{1}{2\| a \|} < a, \nabla f >$ Daf = 2‖a‖1 < a, ∇f >
- ○ $D_a f = \frac{1}{\| a \|^2} < a, \nabla f >$ Daf = ‖a‖21 < a, ∇f >

No, the answer is incorrect.

Score: 0

Accepted Answers:

$D_a f = \frac{1}{\| a \|} < a, \nabla f >$ Daf = ‖a‖1 < a, ∇f >

Let $f(x, y) = e^x \sin y$f(x, y) = ex sin y. If ∇f∇f at the point $(ln(2), \pi/4)$(ln(2), π/4) is $(k, k$k, k) where $k \in R$k ∈ R, then find the value of k^2 k2.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 2

1 point

Level 2

1 point

Let $p = -\frac{k}{(\|r\|)^3} r$p = −(‖r‖)3kr, where $r = (x, y, z) \in R^3 \setminus \{(0,0,0)\}, k \in R$r = (x, y, z) ∈ R3 \ {(0, 0, 0)}, k ∈ R.

Amongst the options below, choose those for which the function ff satisfies $\nabla f = p$∇f = p for some $k \in R$k ∈ R.

Consider dot product as the inner product for finding the norm.

- ☐ $f(x, y, z) = \frac{1}{\|r\|}$f(x, y, z) = ‖r‖1
- ☐ $f(x, y, z) = \frac{1}{2\|r\|}$f(x, y, z) = 2‖r‖1
- ☐ $f(x, y, z) = \frac{1}{\|r\|+2}$f(x, y, z) = ‖r‖+21
- ☐ $f(x, y, z) = \|r\|$ f(x, y, z) = ‖r‖

No, the answer is incorrect.

Score: 0

Accepted Answers:

$f(x, y, z) = \frac{1}{\|r\|}$f(x, y, z) = ‖r‖1

$f(x, y, z) = \frac{1}{2\|r\|}$f(x, y, z) = 2‖r‖1

1 point

Let $f(x, y, z) = 4(x^2 + y^2) - z^2$ f(x, y, z) = 4(x2 + y2) − z2.

Choose the correct option(s).

- ☐ ∇f∇f at (1, 0, 2)(1, 0, 2) is (8, 0, −4)(8, 0, −4)
- ☐ ∇f∇f is continuous everywhere in R^3 R3
- ☐ ∇f∇f is not continuous everywhere in R^3 R3
- ☐ ∇f∇f at (1, 0, 2)(1, 0, 2) is (−8, 0, −4)(−8, 0, −4)

No, the answer is incorrect.

Score: 0

Accepted Answers:

∇f∇f at (1, 0, 2)(1, 0, 2) is (8, 0, −4)(8, 0, −4)

∇f∇f is continuous everywhere in R^3 R3

1 point

Consider a function defined as

$$f(x, y) = \begin{cases} \frac{y^3}{x^2 + y^2} & (x, y) \neq (0, 0) \\ 0 & (x, y) = (0, 0) \end{cases}$$ f(x, y) = {x2+y2y30(x, y) = (0, 0)(x, y) = (0, 0).

Choose the correct option(s).

- ☐ $\frac{\partial f}{\partial x} = \frac{-2xy^3}{(x^2 + y^2)^2}$, ∂x∂f = (x2+y2)2−2xy3, for $(x, y) \neq (0, 0)$(x, y) = (0, 0)

☐ $\frac{\partial f}{\partial x} = \frac{2xy^3}{(x^2+y^2)^2}$, ∂x∂f = (x2+y2)22xy3, for $(x, y) \neq (0,0)$(x, y) = (0, 0)

☐ $\frac{\partial f}{\partial y} = \frac{3x^2y^2+y^4}{(x^2+y^2)^2}$∂y∂f = (x2+y2)23x2y2+y4 for $(x, y) \neq (0,0)$(x, y) = (0, 0)

☐ $\frac{\partial f}{\partial y} = \frac{3x^2y^2-y^4}{(x^2+y^2)^2}$∂y∂f = (x2+y2)23x2y2−y4 for $(x, y) \neq (0,0)$(x, y) = (0, 0)

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\frac{\partial f}{\partial x} = \frac{-2xy^3}{(x^2+y^2)^2}$, ∂x∂f = (x2+y2)2−2xy3, for $(x, y) \neq (0,0)$(x, y) = (0, 0)

$\frac{\partial f}{\partial y} = \frac{3x^2y^2+y^4}{(x^2+y^2)^2}$∂y∂f = (x2+y2)23x2y2+y4 for $(x, y) \neq (0,0)$(x, y) = (0, 0)

1 point

Consider the function definition given in Q6 above. Choose the correct option(s).

☐ $\frac{\partial f}{\partial x}$∂x∂f is continuous at (0, 0)(0, 0)

☐ $\frac{\partial f}{\partial x}$∂x∂f is not continuous at (0, 0)(0, 0)

☐ $\frac{\partial f}{\partial y}$∂y∂f is not continuous at (0, 0)(0, 0)

☐ ∇f∇f is discontinuous at the origin (0, 0)(0, 0)

☐ Directional derivative of ff in the direction of a vector $a \in R^2$a ∈ R2 at (0, 0)(0, 0) can be obtained using the gradient of ff

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\frac{\partial f}{\partial x}$∂x∂f is not continuous at (0, 0)(0, 0)

$\frac{\partial f}{\partial y}$∂y∂f is not continuous at (0, 0)(0, 0)

∇f∇f is discontinuous at the origin (0, 0)(0, 0)

1 point

Consider a function defined as

$$f(x,y) = \begin{cases} (x^2+y^2)\sin\left(\frac{1}{\sqrt{x^2+y^2}}\right) & (x,y) \neq (0,0) \\ 0 & (x,y) = (0,0) \end{cases}$$ f(x, y) = ⌊ . { . ⌈(x2 + y2) sin (x2+y2

√

1)0(x, y) = (0, 0)(x, y) = (0, 0).

Choose the correct option(s).

☐ $\frac{\partial f}{\partial y}(0,0) = 0$∂y∂f(0, 0) = 0

☐ $\frac{\partial f}{\partial x}(0,0) = 0$∂x∂f(0, 0) = 0

☐ $\frac{\partial f}{\partial x} = 0$∂x∂f = 0

☐

$\frac{\partial f}{\partial y} = 0\partial y\partial f = 0$

☐ None of the above

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\frac{\partial f}{\partial y}(0,0) = 0\partial y\partial f(0,0) = 0$

$\frac{\partial f}{\partial x}(0,0) = 0\partial x\partial f(0,0) = 0$

1 point

Consider the function definition given in Q8 above. Choose the correct option(s) with respect to the partial derivative of ff. Note $sign(x) = \pm 1$sign(x) = ±1 depending on the sign of xx.

☐

$\frac{\partial f}{\partial x}(x,0) = 2x\sin(\frac{1}{|x|}) - sign(x)\cos(\frac{1}{|x|})\partial x\partial f(x,0) = 2x\sin(|x|1) - \text{sign(x)}\cos(|x|1)$ for $x \neq 0$ x = 0

☐

$\frac{\partial f}{\partial x}(x,0) = 2x\sin(\frac{1}{|x|}) - sign(x)\cos(\frac{1}{|x|})\partial x\partial f(x,0) = 2x\sin(|x|1) - \text{sign(x)}\cos(|x|1)$

☐

$\frac{\partial f}{\partial x}(x,0) = 2x\sin(\frac{1}{|x|}) - \cos(\frac{1}{|x|})\partial x\partial f(x,0) = 2x\sin(|x|1) - \cos(|x|1)$ for $x \neq 0$x = 0

☐

$\frac{\partial f}{\partial y}(0,y) = 2y\sin(\frac{1}{|y|}) - sign(y)\cos(\frac{1}{|y|})\partial y\partial f(0,y) = 2y\sin(|y|1) - \text{sign(y)}\cos(|y|1)$ for $y \neq 0$ y = 0

☐

$\frac{\partial f}{\partial y}(0,y) = 2y\sin(\frac{1}{|y|}) - sign(y)\cos(\frac{1}{|y|})\partial y\partial f(0,y) = 2y\sin(|y|1) - \text{sign(y)}\cos(|y|1)$

☐

$\frac{\partial f}{\partial y}(0,y) = 2y\sin(\frac{1}{|y|}) - \cos(\frac{1}{|y|})\partial y\partial f(0,y) = 2y\sin(|y|1) - \cos(|y|1)$ for $y \neq 0$y = 0

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\frac{\partial f}{\partial x}(x,0) = 2x\sin(\frac{1}{|x|}) - sign(x)\cos(\frac{1}{|x|})\partial x\partial f(x,0) = 2x\sin(|x|1) - \text{sign(x)}\cos(|x|1)$ for $x \neq 0$ x = 0

$\frac{\partial f}{\partial y}(0,y) = 2y\sin(\frac{1}{|y|}) - sign(y)\cos(\frac{1}{|y|})\partial y\partial f(0,y) = 2y\sin(|y|1) - \text{sign(y)}\cos(|y|1)$ for $y \neq 0$ y = 0

1 point

Consider the function definition given in Q8 above. Choose the correct option.

☐

$\nabla f\nabla$f is continuous everywhere

☐

$\nabla f\nabla$f is discontinuous at the origin $(0,0)(0,0)$

☐

$\frac{\partial f}{\partial x}\partial x\partial$f is continuous at the origin but $\frac{\partial f}{\partial y}\partial y\partial$f is discontinuous at the origin

☐

$\frac{\partial f}{\partial y}\partial y\partial$f is continuous at the origin but $\frac{\partial f}{\partial x}\partial x\partial$f is discontinuous at the origin

☐

Directional derivative of ff in the direction of a vector $a \in R^2$ a ∈ R2 at $(0,0)(0,0)$ cannot be obtained using the gradient of ff

No, the answer is incorrect.

Score: 0

Accepted Answers:

∇f ∇f is discontinuous at the origin $(0, 0)$$(0, 0)$

Directional derivative of f f in the direction of a vector $a \in R^2$ a $\in$ R2 at $(0, 0)$$(0, 0)$ cannot be obtained using the gradient of f f

Check Answers

Your score is: 0/10